

THE CENTRAL LIMIT THEOREM

LAST TIME:

$\{X_i\}$ I.I.D. RANDOM VARIABLES  MEAN μ
VARIANCE σ^2

$$S_n = \sum_{i=1}^n X_i$$

$$M_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$E(S_n) = n\mu$$

$$E(M_n) = \mu$$

$$\text{Var}(S_n) = n\sigma^2$$

$$\text{Var}(M_n) = \frac{1}{n} \sigma^2$$

ALSO THE PDF OF THE SUM (CONVOLUTION)

CONSIDER

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

$$E(Z_n) = 0$$

$$\text{Var}(Z_n) = 1$$

$$\begin{aligned}\text{Var}(Z_n) &= \text{Var}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}}\right) \\ &= \frac{1}{\sigma^2 n} \text{Var}(S_n - n\mu) \\ &= \frac{1}{\sigma^2 n} \text{Var}(S_n) \\ &= \frac{1}{\sigma^2 n} n\sigma^2 = 1.\end{aligned}$$

FOR EVERY n .



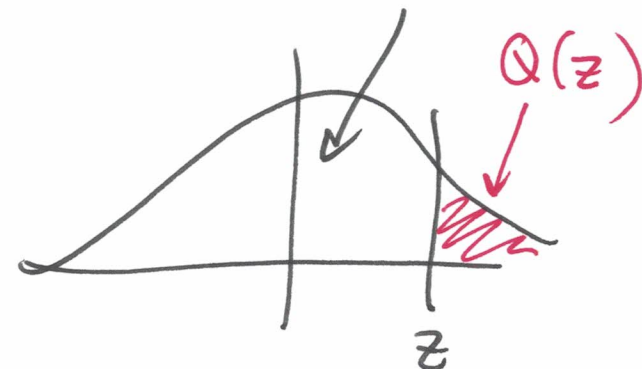
CENTRAL LIMIT THEOREM:

$Z_n \longrightarrow$ GAUSSIAN WITH $\mu=0, \sigma^2=1$!

REGARDLESS OF UNDERLYING PDF OF X_i !

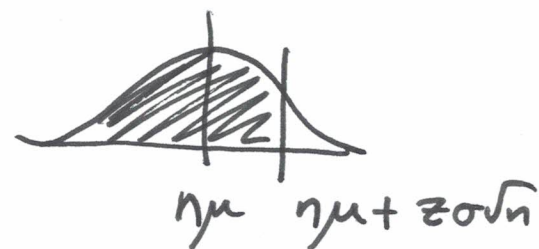
VERSIONS OF THE C.L.T. USEFUL FOR $\phi(z)$
 PROBLEMS / CRIB SHEET:

$$P(Z_n \leq z) \approx \phi(z)$$

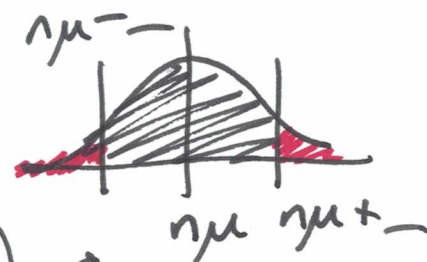


$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) \approx \phi(z) = 1 - Q(z)$$

$$P(S_n \leq z\sigma\sqrt{n} + n\mu) \approx \phi(z)$$

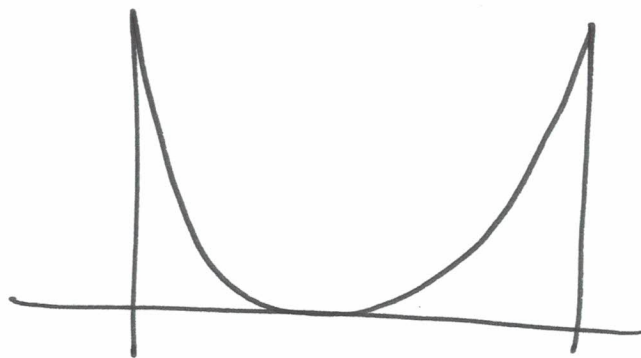
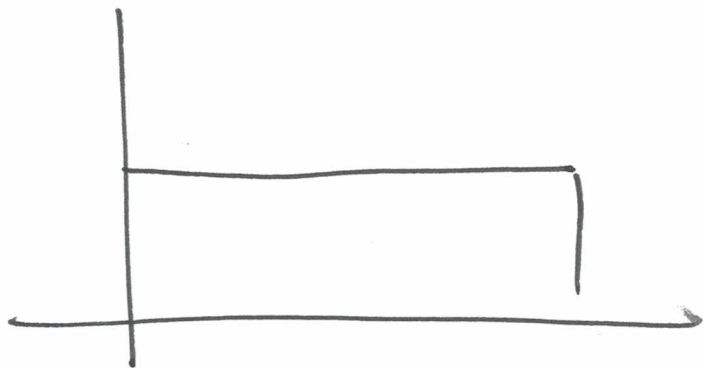
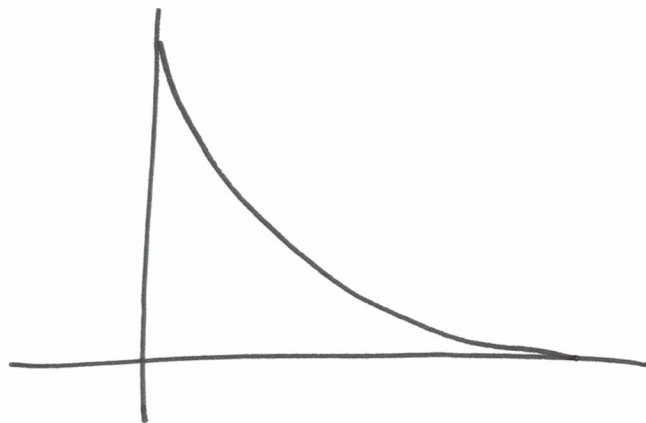
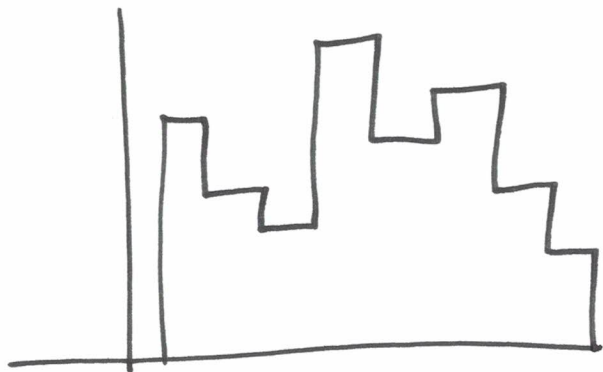


$$P(M_n \leq \frac{z\sigma}{\sqrt{n}} + \mu) \approx \phi(z)$$

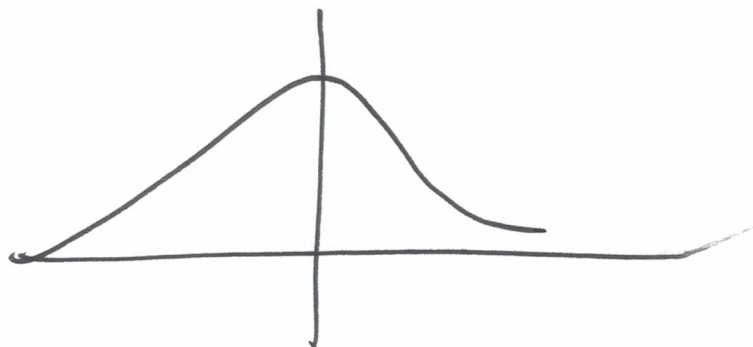


$$P(|M_n - \mu| \leq \frac{z\sigma}{\sqrt{n}}) \approx 1 - 2Q(z)$$

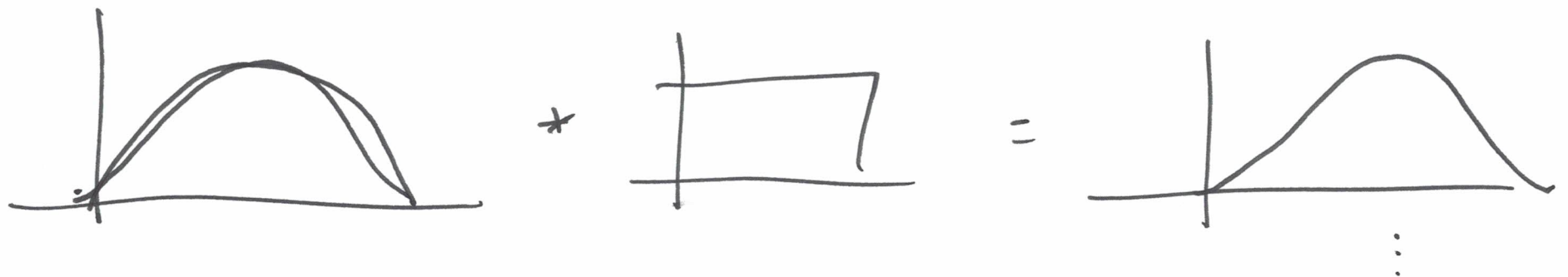
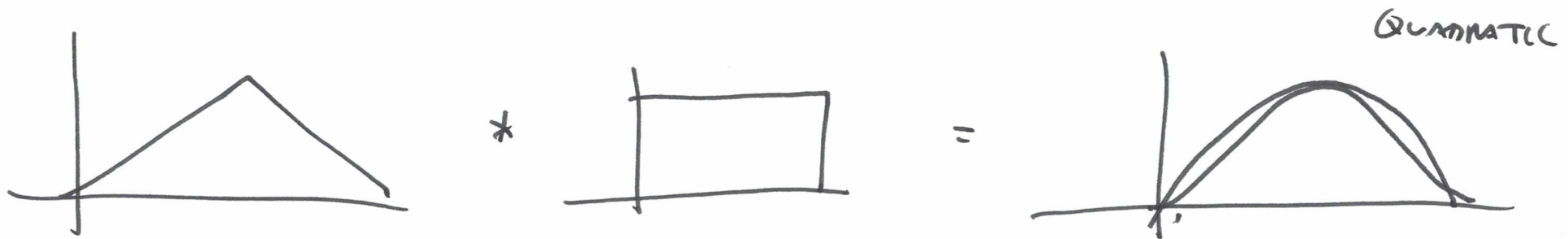
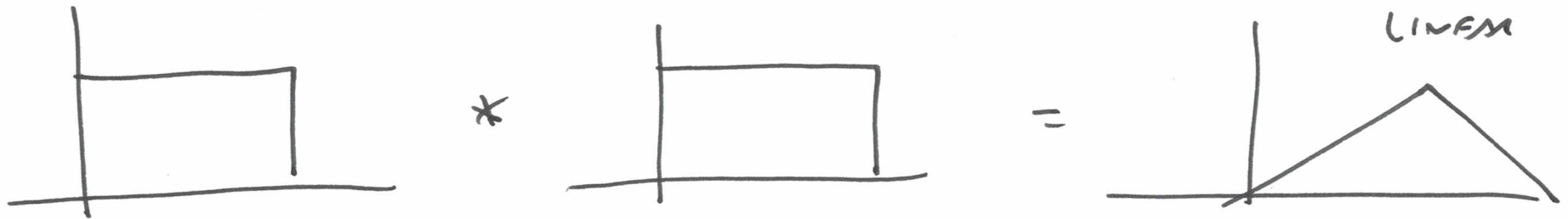
$$P(|S_n - n\mu| \leq z\sigma\sqrt{n}) \approx 1 - 2Q(z)$$

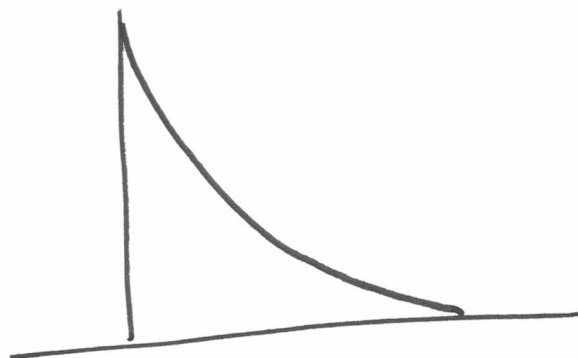


SUM AND
NORMALIZE
IID SAMPLES

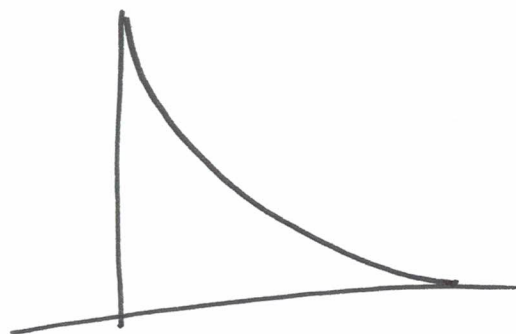


THE ACTUAL PDF IS THE
 REPEATED CONVOLUTION OF THE
 SOURCE PDF AGAIN AND AGAIN.

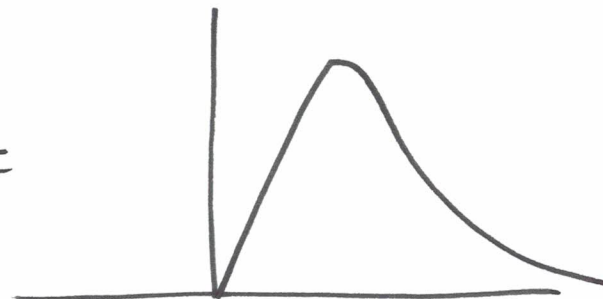




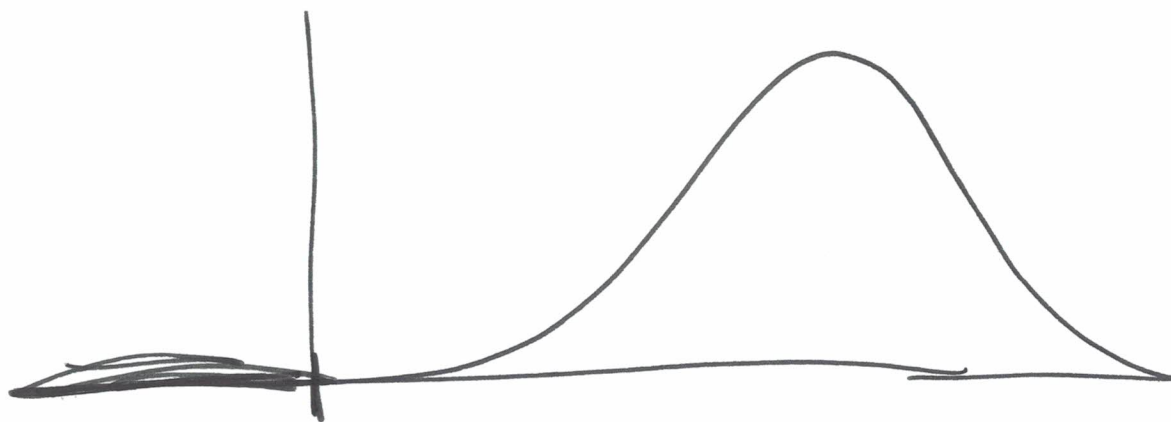
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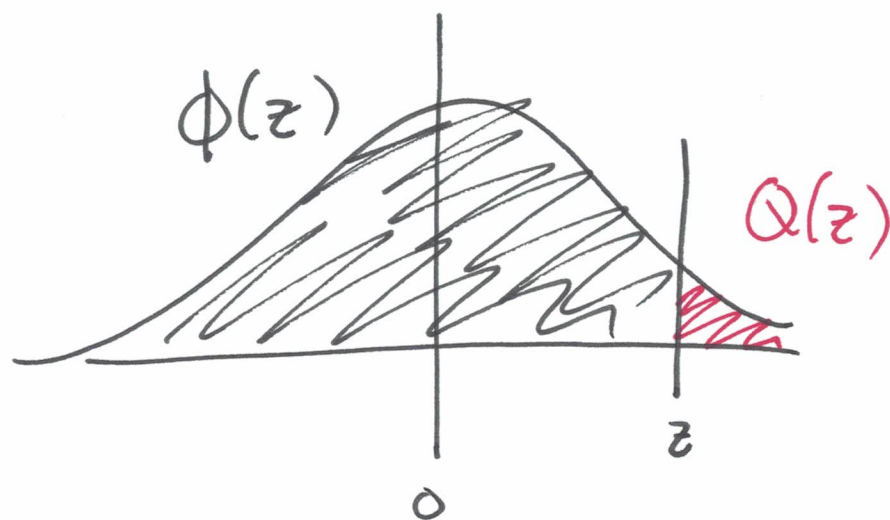


ACTUAL THEOREM:

$$\lim_{n \rightarrow \infty} P(\bar{Z}_n \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{x^2}{2}} dx$$

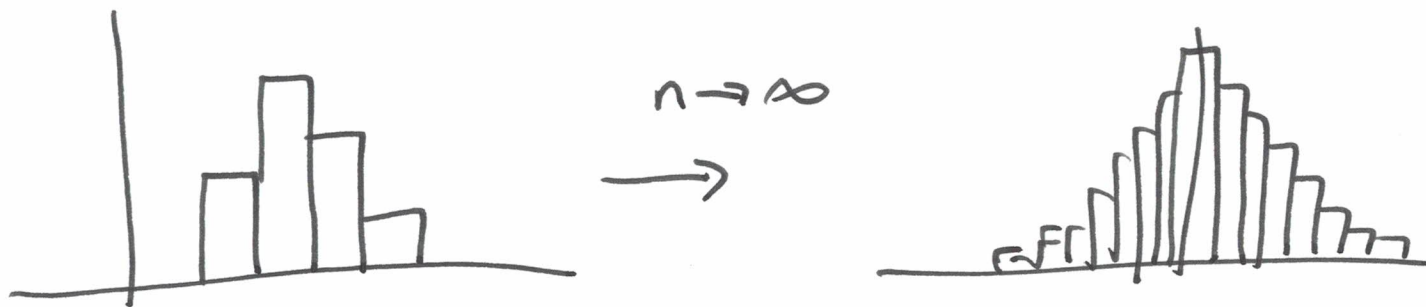
CDF of $\bar{Z}_n$

||
 $\phi(z)$
GAUSSIAN $\mu=0$
 $\sigma=1$



CDF
||
 $1 - Q(z)$

THIS EVEN WORKS FOR DISCRETE R.V.S:



EXAMPLE: BINOMIAL DISTRIBUTION =
SUM OF MANY 0/1 BERNOULLI R.V.
ALREADY LOOKED LIKE A GAUSSIAN!

EX PDF OF POKEMON WEIGHT:

$$f_X(x) = \begin{cases} \frac{1}{36}x + \frac{1}{12} & x \in [0, 6] \\ 0 & \text{ELSE.} \end{cases}$$

ESTIMATE $P(100 \text{ POKEMON WEIGHT} \in [320, 380] \text{ POUNDS})$.
" 350 ± 30 .

WE CAN COMPUTE

$$E(X) = \int_0^6 x \left(\frac{1}{36}x + \frac{1}{12} \right) dx = \frac{7}{2} = \mu$$

$$\text{Var}(X) = \frac{11}{4} = \sigma^2.$$

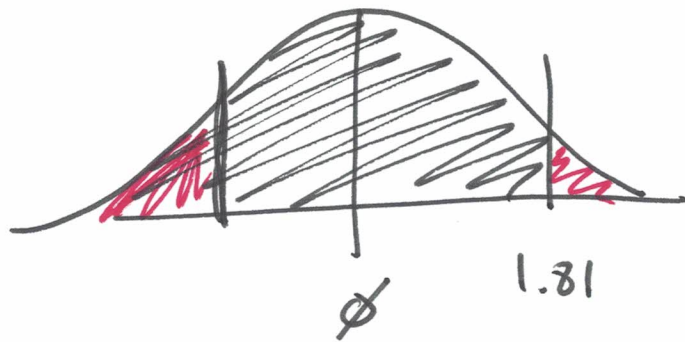
$$100 \text{ POKEMON: } E(S_{100}) = 100 \left(\frac{7}{2} \right) = 350$$

$$P(|S_{100} - \underset{\substack{\text{"} \\ n\mu}}{350}| \leq 30) = ? \quad 1 - 2Q(z)$$

$\hookrightarrow z\sigma\sqrt{n}$

$$z\sigma\sqrt{n} = 30$$

$$z\sqrt{\frac{11}{4}}\sqrt{100} = 30 \quad z = 1.81$$



$$1 - 2Q(1.81) = 1 - 2(0.035) = \boxed{0.93}.$$

HEIGHT OF POKEMON: UNIFORM ON

$[20.5, 30.5]$ INCHES

$P(\text{STACK OF 45 POKEMON CAN REACH THE TOP OF AN 97-FOOT HIGH BUILDING.})$

$$P(S_{45} \geq 97) = Q(z)$$

$$n = 45$$

$$\mu = 25.5$$

$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{10^2}{12}$$

$$8.33$$

$$\sigma = \sqrt{8.33} = 2.89$$

$$97 \cdot 12 = z \cdot \cancel{\sqrt{8.33}} \cdot \sqrt{45} + (45)(25.5)$$

2.89

$$z = 0.852$$

$$Q(z) = 0.197$$

DIFFERENT USAGE: CONFIDENCE INTERVALS

$$P\left(\underset{\substack{\text{LOWER} \\ \text{BOUND}}}{l(x)} \leq \underset{\substack{\uparrow \\ \text{TRUE} \\ \text{MEAN}}}{\mu} \leq \underset{\substack{\text{UPPER} \\ \text{BOUND}}}{u(x)}\right) = 1 - \alpha$$

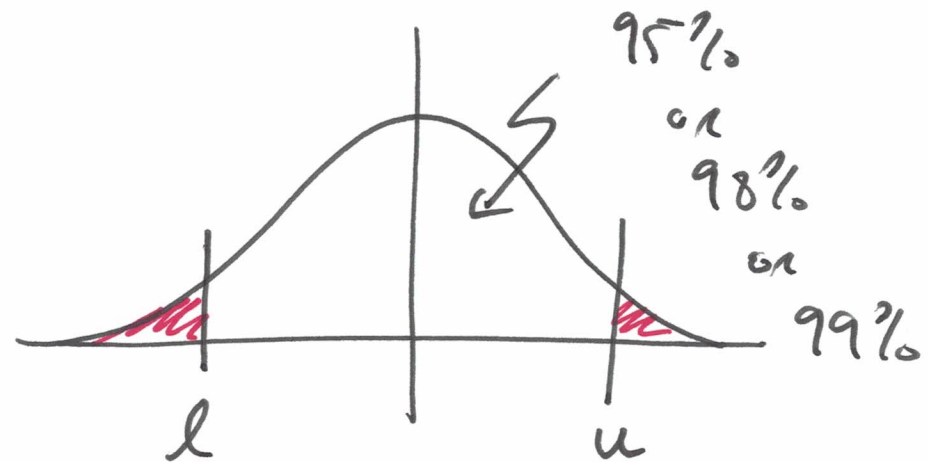
THIS IS THE $(1 - \alpha) * 100\%$
CONFIDENCE INTERVAL FOR ESTIMATING THE
MEAN.

$$99\%$$

"

$$1 - 2Q(z)$$

"



$$P\left(-z \leq \frac{M_n - \mu}{\sigma\sqrt{n}} \leq z\right)$$

$$P\left(\mu \in \left[M_n - \frac{z\sigma}{\sqrt{n}}, M_n + \frac{z\sigma}{\sqrt{n}}\right]\right)$$

THIS INTERVAL ~~IS~~ CONTAINS THE
 TRUE VALUE μ WITH
 PROBABILITY $1 - 2Q(z)$
 (A BIG NUMBER)

EX

WEIGHT OF A FROZEN TURKEY IS

GAUSSIAN R.V. WITH MEAN μ

VARIANCE 4. WEIGH 37 TURKEYS

AND SUM OF WEIGHTS IS ~~32~~ 892

POUNDS. COMPUTE THE 98% CONFIDENCE

INTERVAL FOR THE MEAN.

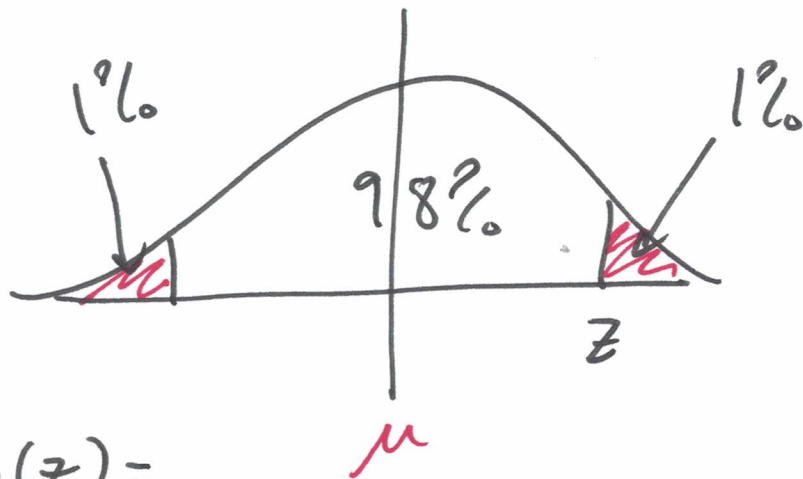
$$z = 2.326$$

$\nearrow$

$$Q(z) = 0.01?$$

WE WILL SUPPLY
KEY INVERSE Z
VALUES.

$$Q(\bar{z}) = \frac{\alpha}{2}$$



$$Q(z) =$$

$$Q\left(\frac{1-0.98}{2}\right) = 0.01.$$

$$M_n = \frac{892}{37} = 24.1$$

SAMPLE
MEAN

$$\left[\underline{24.1} \pm \left[-\frac{z\sigma}{\sqrt{n}}, +\frac{z\sigma}{\sqrt{n}} \right] \right]$$

↓

$$\frac{(2.326)(2)}{\sqrt{37}}$$

"

$$[23.3339, 24.8661]$$

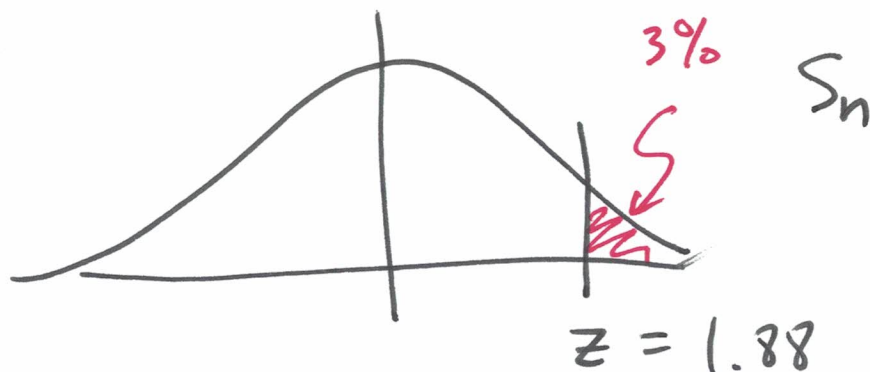
Ex

$X = \# \text{ NUMBER OF ARROWS / HOUR}$

EXPONENTIAL WITH $\mu = 3$.

$$\lambda = \frac{1}{3}$$

HOW MANY ARROWS TO BRING TO 24-HOUR
COMBAT TO BE 97% SURE OF
NOT RUNNING OUT?



$$\boxed{\mu = 3 = \frac{1}{\lambda}}$$

$$\sigma^2 = \frac{1}{\lambda^2}$$

$$\boxed{\sigma = \frac{1}{\lambda} = 3.}$$

SOLVE FOR THIS

$$P(S_n \geq \boxed{z\sigma\sqrt{n} + n\mu}) \leq 0.03$$

S_{24}

$$(\underset{1.88}{\cancel{0.88}})(3)\sqrt{24} + 24 \cdot 3 = 99.6$$

$\leadsto$ 100 arrows

FOR OTHER PROBLEMS (LIKE
UNKNOWN μ , σ , ETC.)

THERE ARE OTHER KINDS OF Q-FUNCTION
LIKE LOOKUP TABLES.

t-TEST

F-TEST...